

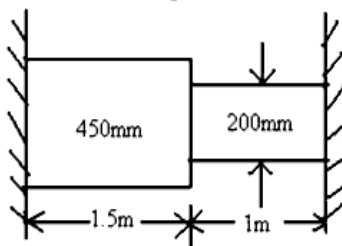
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12

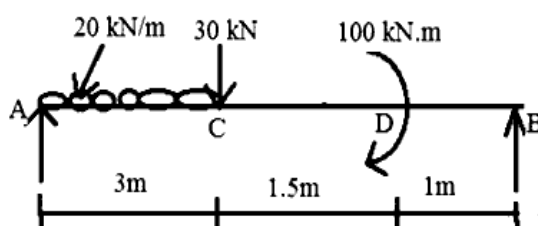
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| <p>A) A M.S bar of 50mm square in size and 150mm long is subjected to an axial thrust of 200kN. Half the lateral strain is prevented by the application of uniform external pressure of certain intensity. If $E = 200 \text{ GPa}$ and Poisson's ratio 0.3. Calculate the change in the length of the bar.</p> | CO1 | 6 |
| <p>B) Write the relationships between bulk modulus, rigidity modulus, young's modulus and poisson's ratio.</p> | CO1 | 6 |
| <p>C) Define stress and strain. What are the different types of stresses and strains? Define the terms: Elasticity, Elastic limit, Young's Modulus and Modulus of rigidity.</p> | CO1 | 6 |

A bar of non-uniform diameter, as shown in figure is rigidly fixed. There is no expansion of the ends and there is no stress in the bar at a temperature of 22°C . If the temperature of the bar be raised to 45°C , find the forces applied by the rigid walls on the bar. MOE and coefficient of thermal expansion for the materials are 200 GN/m^2 and $11.7 \times 10^{-6}/^\circ\text{C}$ respectively. Assume no lateral buckling of the bar.



Q.2 Solve Any Two of the following.	12
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| <p>A) Draw the S.F. & B.M. diagrams for simply supported beam of length L carrying a point load W at its middle point.</p> | CO2 | 6 |
| <p>B) What do you mean by section modulus? Find an expression for section modulus for rectangular, circular & hollow circular sections.</p> | CO2 | 6 |
| <p>C) Draw shear force and bending moment diagram for the beam shown in Fig.</p> | CO2 | 6 |



Q. 3	Solve Any Two of the following.		12
A)	Draw the variation of shear stress for a Tee section? Sketch the shear stress distribution for a circular section. Indicate also the layer at which maximum stress occurs?	CO2	6
B)	A simple beam of span 10m carries a udl of 3kN/m. The section of the beam is a T having a flange of 125x125mm and web 25x175mm. For the critical section obtain the shear stress at the neutral axis and at the junction of flange and the web. Also draw the shear stress distribution across the section.	CO2	6
C)	A beam of channel section 120x60mm has a uniform thickness of 15mm. Draw the shear stress distribution for a vertical section where the shear force is 50kN. Find the ratio between the maximum and mean shear stress.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Derive the Euler's equation for column with two ends fixed.	CO3	6
B)	What are the assumptions and limitations of Euler's theory for long columns?	CO3	6
C)	A slender pin ended aluminium column 2.0 m long and of circular cross section it to have an outside diameter of 50 mm. Calculate the necessary internal diameter to prevent failure by buckling if the actual load applied is 12kN and the critical load applied is twice the actual load. Take E for aluminium as 70 GN/m ²	CO3	6
Q. 5	Solve Any Two of the following.		12
A)	Explain maximum principal strain theory.	CO4	6
B)	Define the terms: Principal planes and Principal stresses. Also explain their uses. Draw the Mohr's circle for a state of pure shear and indicate the principal stresses.	CO4	6
C)	A body is subjected to direct stresses in two mutually perpendicular directions accompanied by a simple shear stress. Draw the Mohr's circle of stresses and explain how you will obtain the principal stresses and principal planes.	CO4	6

END